

RELIABILITY ENGINEERING

Reliability Modelling

Concepts, Formulas, and Worked System Examples

Failure Rate • MTTF & MTBF • Series & Parallel Systems • Case Studies

What is Reliability?

Definition

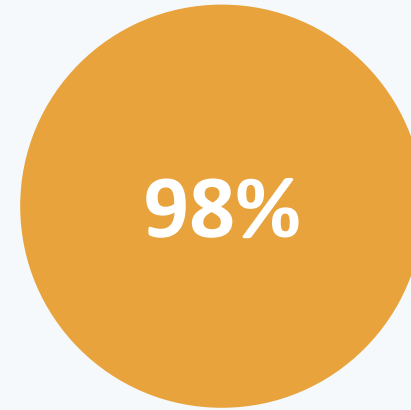
The probability that a system or component performs its intended function without failure for a specified period under stated operating conditions.

$$0 \leq R(t) \leq 1$$

Reliability is always expressed as a probability

Example

A mobile phone has a reliability of 0.98 over one year — a 98% probability it works properly for that year.



$R(t)$

Applies to any system: phones, engines, servers, medical devices — always a value between 0 and 1.

Why Reliability Matters



Aircraft engines



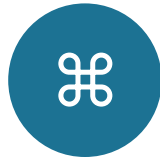
Medical equipment



IoT sensors



Communication networks



Data centers



Manufacturing machines

If a hospital ventilator fails during surgery, the consequences can be severe — reliability analysis helps engineers design systems that minimize such risks.

2. CORE METRICS

Failure Rate (λ)

How frequently a component fails per unit time — measured in failures/hour, failures/day, or failures/year.

Formula

$$\lambda = \frac{\text{Number of Failures}}{\text{Operating Time}}$$

Example

A machine operates for 20,000 hours and fails 10 times.

$$\lambda = 10 / 20,000 = 0.0005 \text{ failures/hour}$$

Interpretation: a smaller λ means higher reliability.



Smaller $\lambda \rightarrow$ Higher Reliability

2. CORE METRICS

MTTF and MTBF



MTTF — Mean Time To Failure

Non-repairable systems: LED bulb, fuse, disposable battery

$MTTF = 1 / \lambda$

$\lambda = 0.002 \rightarrow MTTF = 1/0.002 = 500 \text{ hours}$



MTBF — Mean Time Between Failures

Repairable systems: car, computer server, machine, robot

$MTBF = \text{Operating Time} / \text{Number of Failures}$

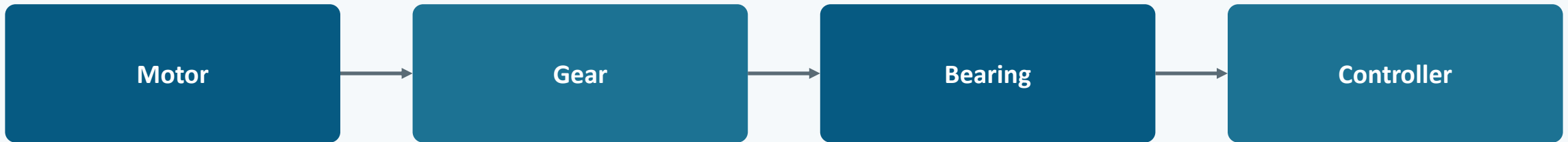
$12,000 \text{ hrs}, 6 \text{ failures} \rightarrow MTBF = 12,000/6 = 2,000 \text{ hours}$

Key Difference

	MTTF	MTBF
Applies to	Non-repairable	Repairable
Measures	Lifetime	Time between failures
Example	Light bulb	Machine

Reliability of Series Systems

If any single component fails, the entire system fails — e.g. Motor → Gear → Bearing → Controller.



A single failure anywhere breaks the whole chain

Formula

$$R = R_1 \times R_2 \times R_3 \times \dots \times R_n$$

Example

Motor 0.98 · Gear 0.95 · Bearing 0.97 · Controller 0.99

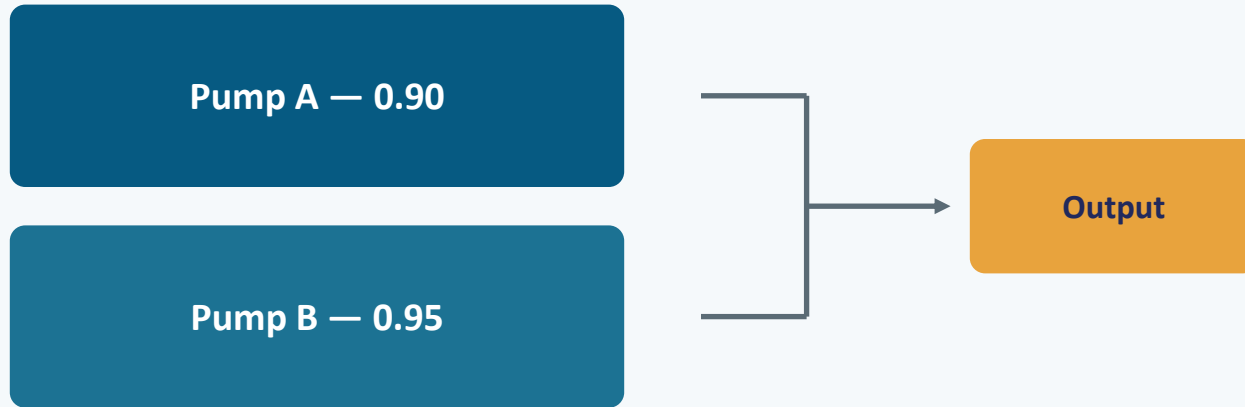
$$R = 0.98 \times 0.95 \times 0.97 \times 0.99 \approx 0.894$$

89.4% — adding more series components generally reduces overall reliability.

3. SYSTEM CONFIGURATIONS

Reliability of Parallel Systems

Redundancy: only one unit needs to work. If Pump A fails, Pump B continues.



Formula

$$R = 1 - (1 - R_1)(1 - R_2)$$

$$R = 1 - (0.10)(0.05) = 0.995$$

99.5% — far higher than either pump alone.



Parallel (redundant) systems improve overall reliability — the more independent paths, the more resilient the system.

4. PRACTICE

Worked Problem: Series Machine

A production machine has three components connected in series.

Component	Reliability
Motor	0.97
Gearbox	0.95
Bearing	0.98

Overall Reliability

$$R = 0.97 \times 0.95 \times 0.98 = 0.903$$

Failure Probability

$$F = 1 - R = 1 - 0.903 = 0.097$$

90.3%

Reliability

9.7%

Failure Probability

Case Study: ATM Machine

An ATM requires its power supply, CPU, card reader, and cash dispenser all to function — a series system.

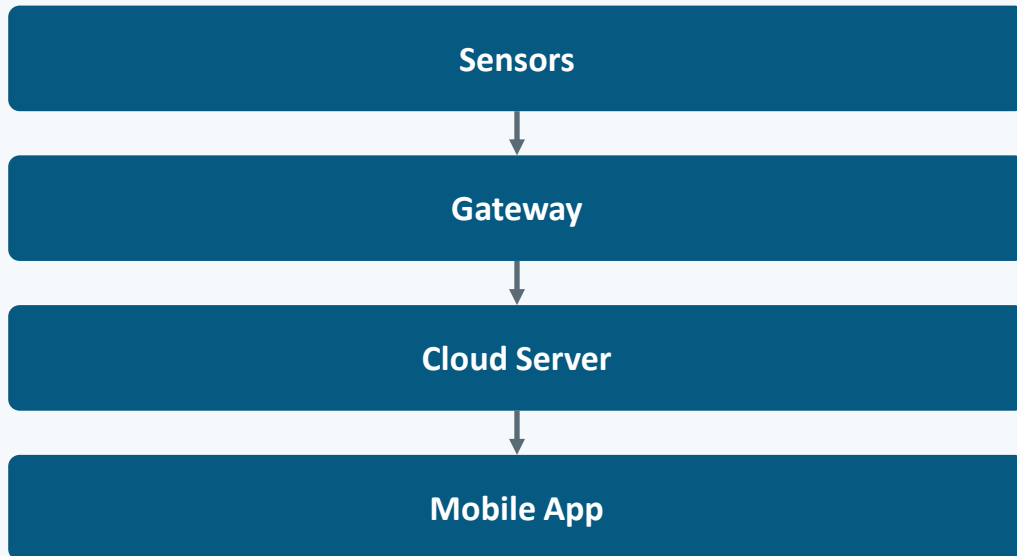


$$R = 0.99 \times 0.98 \times 0.97 \times 0.96 = 0.903 \text{ — only 90.3\% reliable}$$

Discussion question: which component should be improved first to raise the ATM's overall reliability?

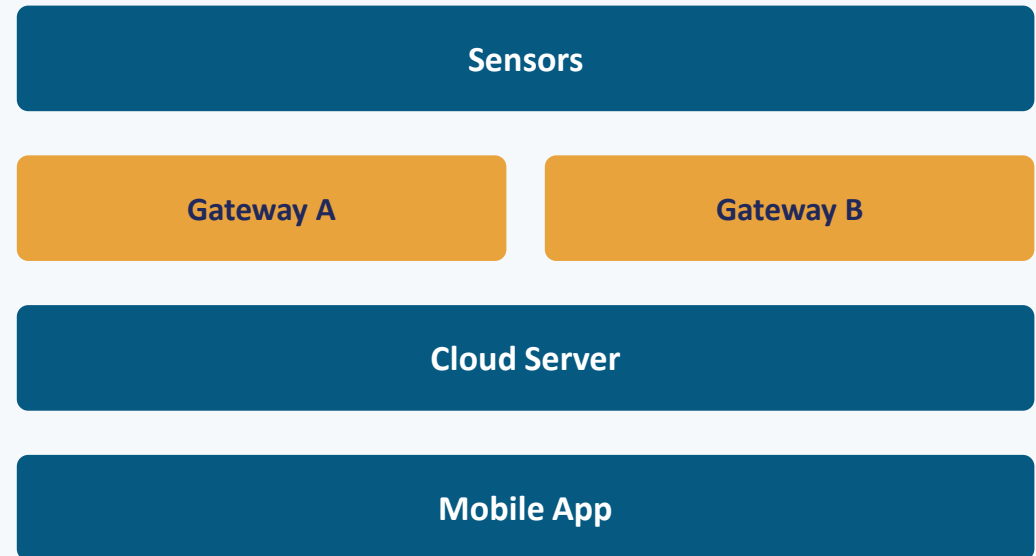
Case Study: IoT Monitoring System

Baseline: series chain



If the gateway fails, no data reaches the cloud.

Improved: redundant gateway



Only one gateway is needed — a parallel sub-system inside the larger series chain.

Adding a second gateway turns that link into a parallel sub-system, increasing overall reliability of the IoT chain.

Worked Example: Heater – Pump – Turbine System

A system has a heater (R1), two pumps in parallel (R2, R3), and 5 turbines (R4–R8).

- Pump sub-system operates if at least ONE of the two pumps works (parallel)
- Turbine sub-system operates if at least 3 of the 5 turbines work
- Heater, pump, and turbine sub-systems are connected in SERIES — all must work

Failure probabilities between scheduled maintenances

R1	R2	R3	R4	R5	R6	R7	R8
0.05	0.10	0.08	0.20	0.17	0.09	0.15	0.15

Goal: compute the reliability of the heater sub-system, the pump sub-system, the turbine sub-system, and the entire system.

6. EXTENDED EXAMPLE

Heater and Pump Sub-Systems

Reliability = $P[\text{system does not fail between scheduled maintenances}]$. W_i = “component i is working properly”.



Heater Sub-System (series-of-one)

$$\text{Reliability} = 1 - P[\text{fails}] = 1 - 0.05$$

$$= 0.95$$



Pump Sub-System (R2, R3 parallel)

$$\text{Reliability} = 1 - P[W2^c \cap W3^c] = 1 - (0.10)(0.08)$$

$$= 0.992$$

Both sub-systems must work for the system to run — the pump sub-system's redundancy makes it far more reliable than a single pump would be.

6. EXTENDED EXAMPLE

Turbine Sub-System and Final Reliability

The turbine sub-system needs ≥ 3 of 5 turbines working. Summing the mutually exclusive working states gives its reliability.



P[Turbine sub-system works]

*= sum of all mutually exclusive
working-state probabilities*

= 97.33%

Whole System Reliability

$$R = R(\text{heater}) \times R(\text{pumps}) \times R(\text{turbines})$$

$$= 0.95 \times 0.992 \times 0.9733$$

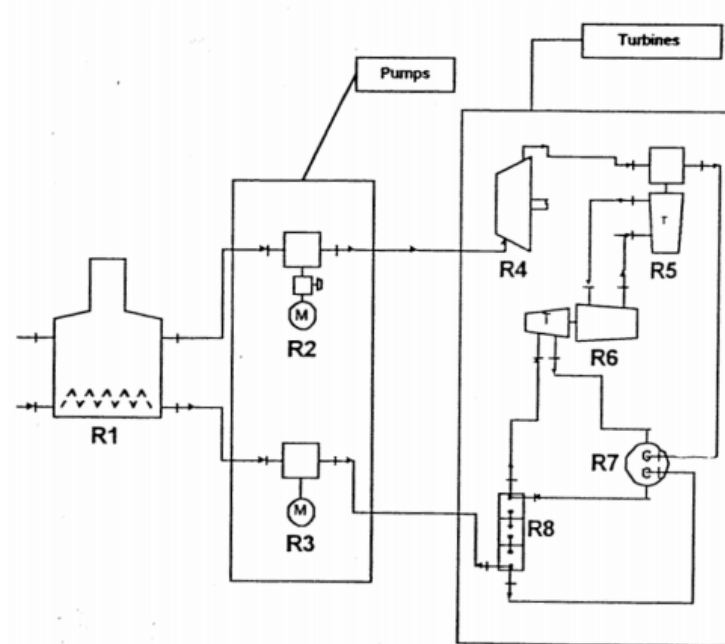
$$= \mathbf{0.9172}$$

91.72% overall system reliability

Discussion — Problem 1.5: Turbines are expensive. How would removing R4 (the least reliable turbine) affect the reliability of the whole system?

Reference Diagram — System & Problem Statement

Example Consider a system composed of a heater (R_1), two pumps (R_2 and R_3), and 5 turbines (R_4 through R_8). The two pumps work in parallel, meaning that the pump sub-system operates if at least one of the pumps operates. The turbine sub-system operates if at least 3 turbines operate. The heater, pump, and turbine sub-systems are connected in series, meaning that they must all properly work for the whole system to perform adequately.



Between two scheduled maintenances, each component may fail independently of the others, with the following probabilities

R1	R2	R3	R4	R5	R6	R7	R8
0.05	0.10	0.08	0.20	0.17	0.09	0.15	0.15

Reference Solution — Reliability Calculations

The reliability of the system is defined as the probability that the system does not fail between scheduled maintenances. We denote by W_i the event “component i is working properly”. To calculate system reliability, we first consider the reliability of each sub-system separately:

- heater sub-system: Reliability = $P[W_1] = 1 - 0.05 = 0.95$

- pump sub-system:

$$\text{Reliability} = 1 - P[W_2^c \cap W_3^c] = 1 - P[W_2^c]P[W_3^c] = 1 - (0.10)(0.08) = 0.992$$

- turbine sub-system: All possible events for which the turbine sub-system is in working conditions and their probabilities are listed below. Since the events are mutually exclusive, the probability of their union (which is the same as the event “the turbine sub-system works properly”) is the sum of their individual probabilities, i.e. 0.9733.

R4	R5	R6	R7	R8	P[W(turbines)]
Works	Works	Works	Works	Works	43.66%
Fails	Works	Works	Works	Works	10.91%
Works	Fails	Works	Works	Works	8.94%
Works	Works	Works	Works	Fails	7.70%
Works	Works	Works	Fails	Works	7.70%
Works	Works	Fails	Works	Works	4.32%
Fails	Fails	Works	Works	Works	2.24%
Fails	Works	Works	Works	Fails	1.93%
Fails	Works	Works	Fails	Works	1.93%
Works	Fails	Works	Works	Fails	1.58%
Works	Fails	Works	Fails	Works	1.58%
Works	Works	Works	Fails	Fails	1.36%
Fails	Works	Fails	Works	Works	1.08%
Works	Fails	Fails	Works	Works	0.88%
Works	Works	Fails	Works	Fails	0.76%
Works	Works	Fails	Fails	Works	0.76%
					97.33%

Finally, the reliability of the entire system is $(0.95)(0.992)(0.9733) = 0.9172$.

Problem 1.5 Turbines are very expensive. Therefore, there are financial incentives to removing one of them. How would removal of turbine R4, which is the least reliable one, affect the reliability of the whole system?